Space-Time-Separable Graph Convolutional Network for Pose Forecasting

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Human Pose Forecasting

Goal:
Model the complex ***spatio-temporal*** dynamics of human poses.
Observe 10 frames (0.4 seconds) to predict body joints for the future 25 frames (1 second).

Our Contributions:
- First ***GCN-only*** model (***1.7%*** parameters w.r.t. best current technique[1])
- First ***factorized-adjacency*** GCN
- STS-GCN improves on the state-of-the-art over ***32%***

Prior Work:
Other approaches encode space and time by separate frameworks. Time is encoded by RNN[1], LSTM[2]. Space is encoded by GCN:
- Separable convolutions have been successful for efficiency and accuracy, e.g. Xception[3].
- Only [1] applies this concept to GCNs, but connecting body joints along a kinematic tree.

Code is available at: https://github.com/FraLuca/STSGCN

STS-GCN

Coordinates are **encoded** by the proposed Space-Time-Separable Graph Convolutional Network. The **decoding** phase is modelled with a TCN.

- **GCN**

Performs a convolution over a graph in which the relationships between nodes are represented by an adjacency matrix.

$$H^{(l+1)} = \sigma(A^{st-(l)} H^{(l)} W^{(l)})$$

- **Our proposed STS-GCN**

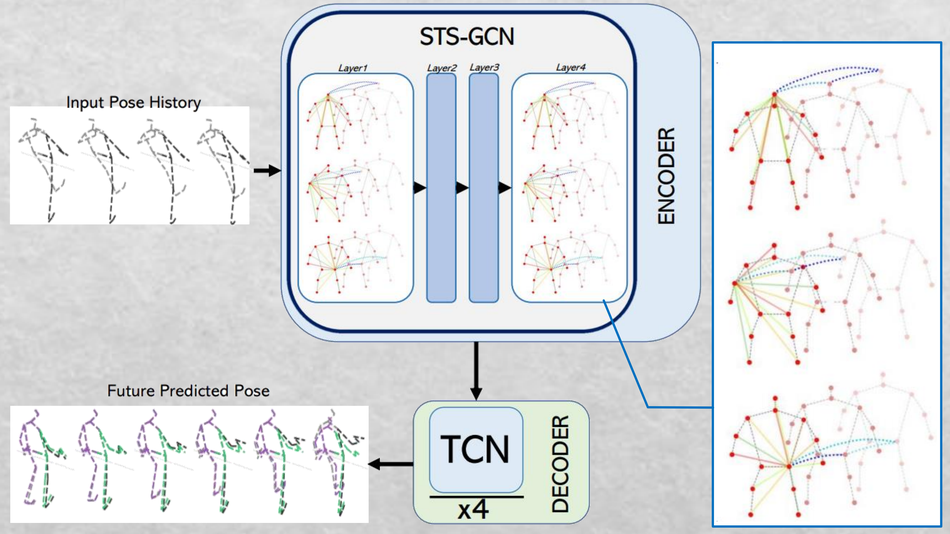
A separable space-time graph convolutional layer l is written as follows:

$$H^{(l+1)} = \sigma(A^{s-(l)} A^{t-(l)} H^{(l)} W^{(l)})$$

We simplified this structure by separating the spatial from the temporal dimension, factorizing the adjacency matrix. This allowed us to have a fraction of the parameters, resulting in a lighter and more efficient model.. $\boldsymbol{A^s}$ and $\boldsymbol{A^t}$ are responsible for the joint-joint and the time-time relations. They are full and trainable matrices.

References

[1] W.Mao et al., History repeats itself: Human motion prediction via motion attention. In *ECCV*, 2020.
[2] Y.Yuan et al., Ego-pose estimation and forecasting as real-time pd control. In *ICCV*, 2019.
[3] F. Chollet. Xception: Deep learning with depthwise separable convolutions. In *CVPR*, 2017.
[4] C.Ionescu et al., Human3.6m: Large scale datasets and predictive methods for 3d human sensing in natural environments. In *TPAMI*, 2014.
[5] N.Mahmood et al., Amass: Archive of motion capture as surface shapes. In *ICCV*, 2019.
[6] T.von Marcard et al., Recovering accurate 3d human pose in the wild using imus and a moving camera. In *ECCV*, 2018.



Quantitative Results

On three complex, recent and large-scale datasets, STS-GCN outperforms the state-of-the-art surpassing the current best technique by over ***32%*** in average at the most difficult long-term predictions, with a reduction of ***98.3%*** of its parameters.

Human 3.6M [4]

msec	560	720	880	1000
ConvSeq2Seq	90.7	104.7	116.7	124.2
LTD-50-25	79.6	93.6	105.2	112.4
LTD-10-25	79.5	94.0	105.6	112.7
LTD-10-10	78.3	93.3	106.0	114.0
DCT-RNN-GCN	77.3	91.8	104.1	112.1
Ours	**50.8**	**60.1**	**68.9**	**75.6**

At least ***34%*** less error

AMASS [5]

msec	560	720	880	1000
ConvSeq2Seq	79.0	87.0	91.5	93.5
LTD-10-10	61.5	75.9	86.2	91.2
LTD-10-25	57.2	65.7	71.3	75.2
DCT-RNN-GCN	51.7	58.6	63.4	67.2
Ours	**31.9**	**38.1**	**42.7**	**45.5**

At least ***32%*** less error

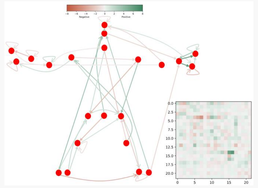
3DPW [6]

msec	560	720	880	1000
ConvSeq2Seq	69.4	77.0	83.6	87.8
LTD-10-10	59.1	69.1	76.5	81.1
LTD-10-25	57.9	65.8	71.5	75.5
DCT-RNN-GCN	56.0	63.6	69.7	73.7
Ours	**30.4**	**35.7**	**39.6**	**42.3**

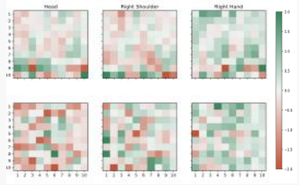
At least ***43%*** less error

Learnt Matrices Visualization

Visualizing the learnt adjacency matrices allows to interpret what the model learns.



Joint-joint edge relations mainly follow the kinematic tree, but also reflect long-term connections or correspondences



The learnt temporal matrix shows information flow from the earlier to the later observed frames

Ablation Study

Ablative variants:
- Separate graph for space and time
- Full non-separable graph
- Separable graph shared across layers
- Separable graph not shared across layers

The proposed **factorized** model has best performance and it is **parameter-efficient**

msec	560	720	880	1000	Parameters
Distinct G^s, G^t	58.7	66.9	75.2	79.9	59.8k
Full G^{st}	53.1	65.6	75.1	82.5	222.9k
Separable G^{s-t} shared	52.5	62.1	69.2	76.9	36.4k
Separable G^{s-t} (proposed)	**50.8**	**60.1**	**68.9**	**75.6**	**57.5k**

Take Home Message

Novelties:
- First ***GCN-only model***
- First network to **factorize** the GCN adjacency matrix

Strengths:
- STS-GCN improves on the state-of-the-art over ***32%***
- Parameters reduction of ***98.3%*** w.r.t. current best technique[1]

Open Research Questions:
- Learnt adjacency matrices have real values and are asymmetric
  - **Signed and directed GCNs** are still relatively **unexplored**
- Currently a spatial GCN
  - **Spectral interpretation** requires new research